LOYOLA UNIVERSITY CHICAGO

Introduction to Computing

MODULE 1: DIGITAL INFORMATION

Outlines

HISTORY OF COMPUTERS

We will talk about the different generations of computers and how they changed the way we solve problems.

CODING AND PROGRAMMING

We will talk about computer programs and how they communicate with the hardware. We talk about operating systems and their components

WORKING WITH BINARY

We will talk about how data is represented in a computer system, and how binary bits build larger units to process, store, and transmit information.

History of Computers

- There are different generations of computing devices.
- Computer generations are characterized by major technological developments that altered the way computers operate.
    - More computational power.
    - More reliability.
    - More efficiency
    - Smaller in size.
    - Cheaper
    - Etc.

First Generation

1940-1956
Vacuum Tubes

Second Generation

1956-1963
Transistors

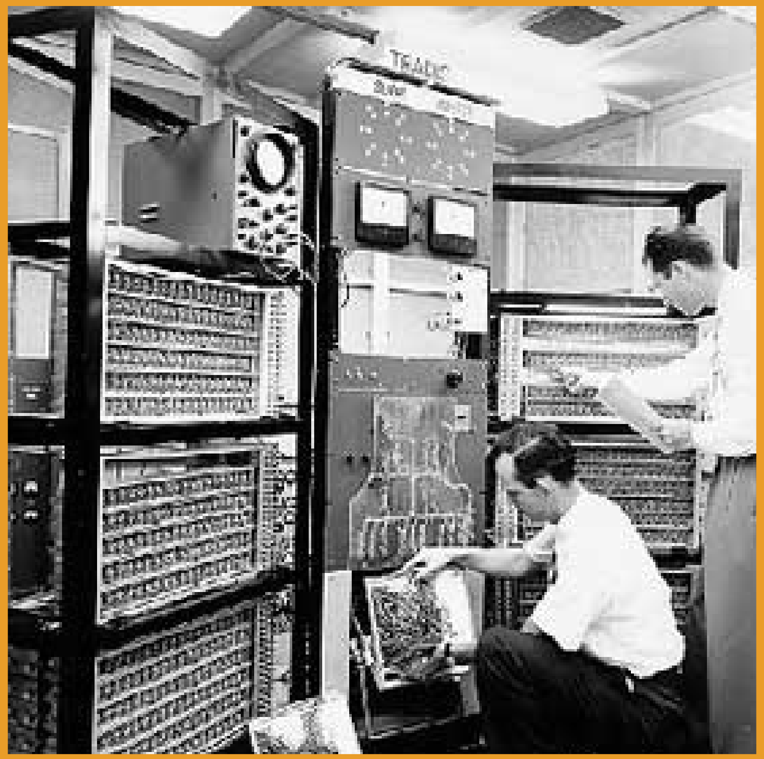
Third Generation

1964-1971
Integrated circuits

Fourth Generation

1971-Present
Microprocessors
VLSI & ULSI

Future Generation



- Vacuum tube technology
- Unreliable
- Machine language only
- Very costly
- Generated a lot of heat
- Slow input and output devices
- Huge size - Non-portable
- Consumed a lot of electricity

- ENIAC
- EDVAC
- UNIVAC
- IBM-701
- IBM-650

- Use of transistors
- More Reliable
- Smaller size
- Less heat
- Consumed less electricity
- Faster
- Still very costly
- Machine & assembly languages

- IBM 1620
- IBM 7094
- CDC 1604
- CDC 3600
- UNIVAC 1108

- Use of IC
- More Reliable
- Smaller size
- Less heat
- Consumed less electricity
- Faster
- Still costly
- High-level languages

- IBM-360 series
- Honeywell-6000 series
- PDP (Personal Data Processor)
- IBM-370/168
- TDC-316

- VLSI - Very Large Scale Integration
- ULSI - Ultra Large Scale Integration
- Smaller size
- Less heat
- Consumed less electricity
- Faster
- Cheaper
- High-level languages

- Desktop
- Laptop
- NoteBook
- UltraBook
- ChromeBook

EXAMPLES

Impact of Innovation

The impact of the innovation of computers and computer science extends far beyond the computer.

MEDICINE

Robotics and sensors are used to enter the body and detect medical conditions that would otherwise have gone unnoticed

ART AND FASHION

Computers are used to build 3D printed clothing, pieces of art, music, patterns and more.

TRANSPORTATION

Computers inform cities on how to best configure train and bus routes, timing of traffic, and the number of lanes needed to reduce congestion.

CODING AND PROGRAMMING

Programs are what allow the hardware to come alive.

This can happen at many levels

- **Operating System:**
  - Communicate directly to the hardware components.
    - e.g., CPU, Memory, and Input/output devices.
  - Used to be very simple (**text terminal**), and it has become very sophisticated.
    - Examples: macOS, Linux, Windows, iOS and Android.
  - Defines file formatting systems to store information.
  - Defines the way programs (applications) access the hardware.
    - This is done using **API** (Application Programming Interface).
  - Provides a set of tools that allow programmers to build applications.
    - These tools are called **SDK**s (Software Development Kits).

SDK

- The SDKs include three basic parts.
    - Pre-built sets of code.
        - Define ways to work with numbers, arithmetic, and the file system.
        - Defines the user interface controls
            - i.e., buttons and windows
        - Defines ways to navigate and use the hardware
            - e.g., keyboard, mouse, touch, motion or other sensors.
    - Programming language.
    - Interpreter or compiler.

THE INTERPRETER

- The interpreter takes written code and translates it to instructions.
- From language programmers understand to language machines understand.
- Operates live (line by line) on script languages (interpreted languages) -
    - Most of these languages use an intermediate representation
        - i.e., combines compiling and interpreting.
- Example: Python and JavaScript

THE COMPILER

- The compiler takes all the program and converts it all to machine code.
- Compiled programs tend to be faster than running a script.
- Multiple levels of optimization can be used.
- Changes to the program require re-compilation.
- Example: C, C++, and Java

Everything you run on a device needs to be translated to the machine language.

Writing a novel, composing music, texting/typing, playing a game, analyzing data, and all other applications (including the OS).

MACHINE LANGUAGE

WORKING WITH BINARY

BINARY SYSTEM

- Binary system has only two possible states.
  - On/Off - True/False - 1/0
- Binary state can be represented with:
  - Boolean value (True or False)
  - Number (1 or 0)
- Data can be stored and communicated using **on** or **off** states.
- Logic gates in microchips use electricity to capture the **on** and **off** signals.
- Storage devices and networks save binary states and communicate them.
- Single state is called a **bit**, which is used to build larger and more complicated data units.

BINARY: UNITS

- Using binary bits as basic unit, data is represented as a string of bits.
- Eight binary digits is called a **byte**, which is a common measure of data.

Bit	Byte	Kilobyte	Megabyte	Gigabyte	Terabyte
	8 bits	1,024 bytes	1,024 kilobytes	1,024 megabytes	1,024 gigabytes

- If your computer has 1-TB, it can store almost **nine trillion bits**.
- Each bit holds a state that is a tiny part of a much larger piece of information.
  - e.g., text, picture, music, movie, or program.

BINARY: ENCODING AND DECODING

- Encoding: The process of converting information (e.g., text or image) to binary.
  - To process, store, or transmit information.
- Decoding: The process of converting binary to information.
  - To reconstitute information in a format we recognize.

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MACHINE LANGUAGE

COMMUNICATING WITH BINARY

DEVELOPING A PROTOCOL

- Protocol: a set of rules that outline the foundation for digital communications and how bits are transmitted..
  - Bit size: number of bits per message.
  - Bit rate: number of bits that are transmitted per second (e.g., 1 bps).
  - Termination
- Example: **0110** What does it mean?
  When to expect the next?

0	Is it a traffic light?
1	Is it red?
1	Is it on?
0	End communication?

DATA COMMUNICATION

- Transmitting data can be performed in different ways.

Radio

Electrical

Optical

RADIO: WI-FI NETWORKS

- Transmit binary states based on alternating the radio waves.
- The process:
    - Encodes to binary.
    - Transmits by varying radio waves.
    - Receives these waves
    - Decodes to binary
- Wi-Fi and radio communications work well for short distances.
    - Devices should be close to the connection hotspot.
- May encounter much interference and noise.
    - When having multiple hotspots.

RADIO: WI-FI NETWORKS

- Hotspots work with unique channels to minimize interference and noise.
- **2.4-GHz** frequency band
  - 14 channels designated for wireless networks.
  - Channels are spaced 5 MHz apart.
  - The protocol requires 25 MHz of channel separation
  - Channels 1, 6, 11 are typically used in the US.
- **5-GHz** frequency band
  - 42 channels designated for wireless networks.
  - Includes four bands: 5.1-GHz, 5.3-GHz, 5.4-GHz, and 5.8-GHz
  - Channels are non-overlapping.

ELECTRICAL NETWORKS

- Send electric pulses through wires/cables that represent binary states
  - For example, ethernet cable.
- This kind of networks makes the foundation of most communication.
  - For example, you may use the Wi-Fi but the hotspot is wired to a larger network.
- There are some limitations when considering long distances.
  - Limitations include speed and reliability.

OPTICAL OR LIGHT NETWORKS

- Send pulses of light through a cable of glass or plastic.
  - The speed of light through the glass:
    - 200,000 kilometers/s (124,000 mile/s)
- Optical cables are costly.

Radio Optical

- These networks are used across continents and oceans.

ALL TOGETHER

Radio, electrical, and optical

We are able to communicate across the globe.

Even outside.

05 MINUTES DISCUSSION

For optical communication, how to encode on and off?

Well, like a light switch.

But what about having multiple zeros in a row to send?

SOME CONCEPTS

- The bit rate for a communication system is very important.
    - Example: with a bit rate 1 bit/s, when not having signal for
        - 1 second -> 0
        - 2 seconds -> 00
        - 3 seconds -> 000

Radio Optical

- Other important concepts are: **bandwith** and **latency**.

Bandwidth

The theoretical maximum that a transmission system can support.

- USB 3 port has **5 gigabits** (Gbps)
- USB 4 port has **40 gigabits** (Gbps)

Latency

The time to get a message from one place to another. Usually, in milliseconds (1/1000 of a second).

NUMBER SYSTEMS

NUMBER SYSTEMS

- **Numerical numbers:**
  - Using decimal digits 0 1 2 3 4 5 6 7 8 9
  - Numbers use a base 10 for multiple digits.
- **Roman numerals:**
  - Using symbols like I, V, X, L, C, and M
  - There can be three consecutive symbols of the same type.
  - I II III IV V VI VII VIII IX X
  - L (50) C(100) D(500) M(1,000)

DECIMAL NUMBER SYSTEM

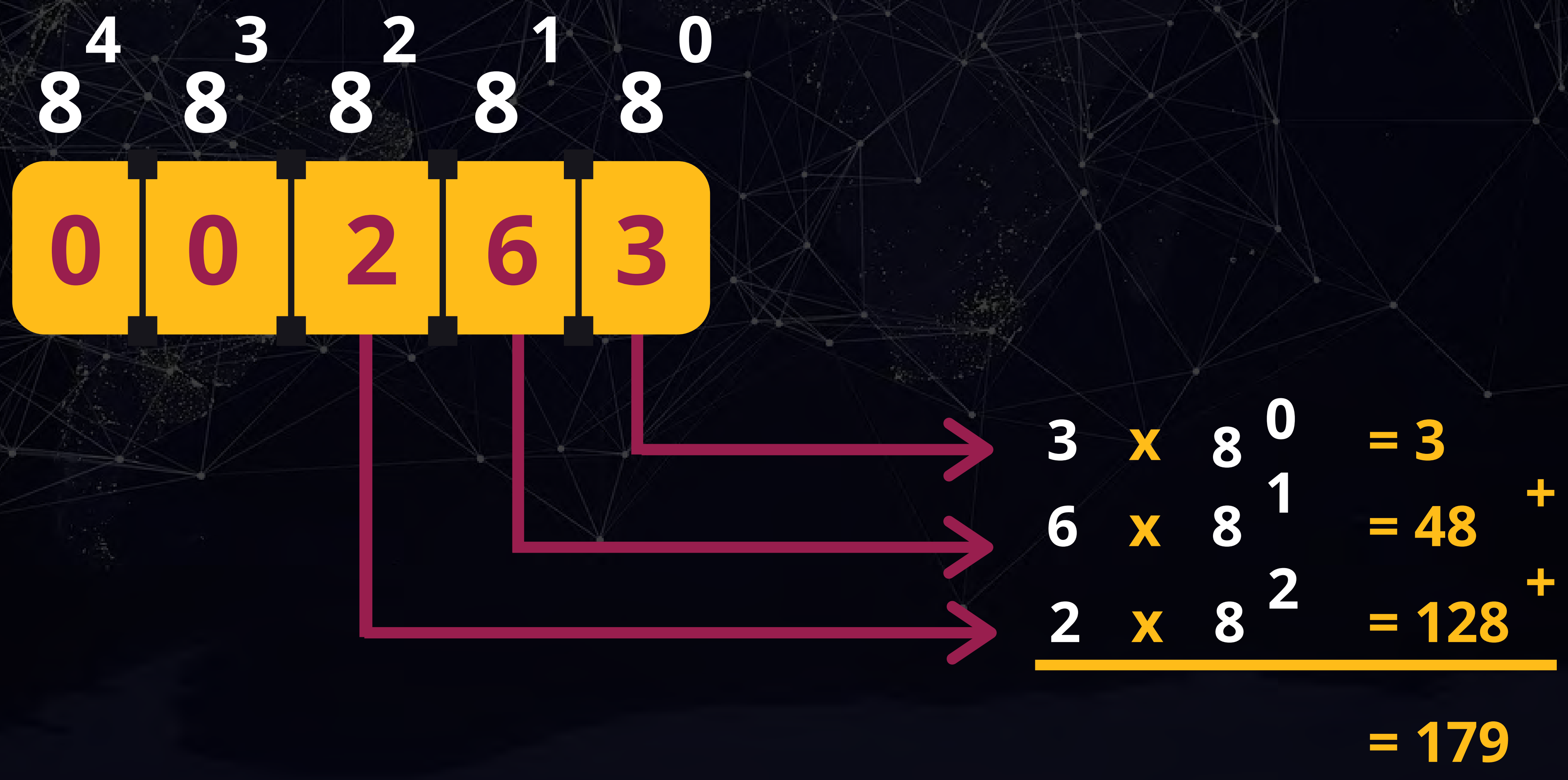
- Numbers use a **base 10** for multiple digits
- When reaching the highest digit and wanting to add one more.
  - Open a left column with one and set the current column to zero.

10^4	10^3	10^2	10^1	10^0
0	0	2	1	5

$$5 \times 10^0 = 5$$
$$1 \times 10^1 = 10 \quad +$$
$$2 \times 10^2 = 200 \quad +$$
$$= 215$$

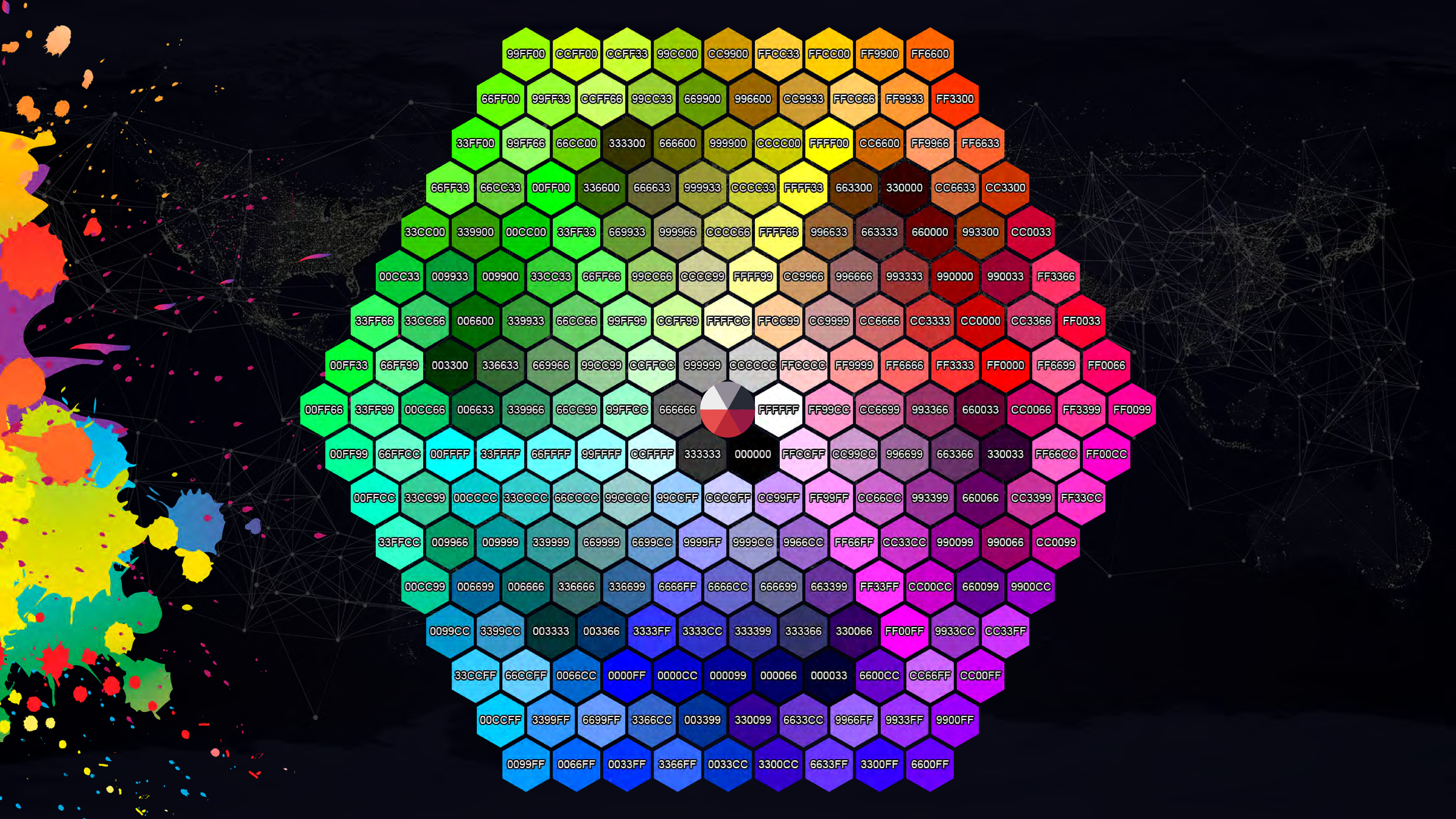
OCTAL NUMBER SYSTEM

- Numbers use a **base 8** for multiple digits



HEXADECIMAL NUMBER SYSTEM

- Numbers use a **base 16** for multiple digits
- After 9, use A, B, C, D, E, and F
- Often used for coding and web design.
  - Colors are represented by three or six hexadecimal numbers.
    - Representing red, green, and blue (**RGB**)


99FF00 CCFF00 CCFF33 99CC00 CC9900 FFCC33 FFCC00 FF9900 FF6600
66FF00 99FF33 CCFF66 99CC33 669900 996600 CC9933 FFCC66 FF9933 FF3300
33FF00 99FF66 66CC00 333300 666600 999900 CCCC00 FFFF00 CC6600 FF9966 FF6633
66FF33 66CC33 00FF00 336600 666633 999933 CCCC33 FFFF33 663300 330000 CC6633 CC3300
33CC00 339900 00CC00 33FF33 669933 999966 CCCC66 FFFF66 996633 663333 660000 993300 CC0033
00CC33 009933 009900 33CC33 66FF66 99CC66 CCCC99 FFFF99 CC9966 996666 993333 990000 990033 FF3366
33FF66 33CC66 006600 339933 66CC66 99FF99 CCFF99 FFFFCC FFCC99 CC9999 CC6666 CC3333 CC0000 CC3366 FF0033
00FF33 66FF99 003300 336633 669966 99CC99 CCFFCC 999999 CCCCCC FFCCCC FF9999 FF6666 FF3333 FF0000 FF6699 FF0066
00FF66 33FF99 00CC66 006633 339966 66CC99 99FFCC 666666 FFFFFF FF99CC CC6699 993366 660033 CC0066 FF3399 FF0099
00FF99 66FFCC 00FFFF 33FFFF 66FFFF 99FFFF CCFFFF 333333 000000 FFCCFF CC99CC 996699 663366 330033 FF66CC FF00CC
00FFCC 33CC99 00CCCC 33CCCC 66CCCC 99CCCC 99CCFF CCCCFF CC99FF FF99FF CC66CC 993399 660066 CC3399 FF33CC
33FFCC 009966 009999 339999 669999 6699CC 9999FF 9999CC 9966CC FF66FF CC33CC 990099 990066 CC0099
00CC99 006699 006666 336666 336699 6666FF 6666CC 666699 663399 FF33FF CC00CC 660099 9900CC
0099CC 3399CC 003333 003366 3333FF 3333CC 333399 333366 330066 FF00FF 9933CC CC33FF
33CCFF 66CCFF 0066CC 0000FF 0000CC 000099 000066 000033 6600CC CC66FF CC00FF
00CCFF 3399FF 6699FF 3366CC 003399 330099 6633CC 9966FF 9933FF 9900FF
0099FF 0066FF 0033FF 3366FF 0033CC 3300CC 6633FF 3300FF 6600FF

HEXADECIMAL NUMBER SYSTEM

16^4	16^3	16^2	16^1	16^0
0	A	3	7	E

$$14 \times 16^0 = 14$$
$$+$$
$$7 \times 16^1 = 112$$
$$+$$
$$3 \times 16^2 = 768$$
$$+$$
$$10 \times 16^3 = 40960$$

$$= 41854$$

BINARY NUMBER SYSTEM

- Numbers use a **base 2** for multiple digits

2^4	2^3	2^2	2^1	2^0
0	1	1	1	0

$$0 \times 2^0 = 0$$
$$+$$
$$1 \times 2^1 = 2$$
$$+$$
$$1 \times 2^2 = 4$$
$$+$$
$$1 \times 2^3 = 8$$

$$= 14$$

VALUES

BIT SIZE AND VALUES

- Binary represents numbers and values in a way the computer understands.
- Two binary digits can store a number with a maximum value of 3.
  - Increasing the digits increases the value we can store.
  - For example, 8-bit, 16-bit, 32-bit, 64-bit numbers
- **8-bit** can store values from 0 to 255.
  - A total options of values = 2^8 = **256**.
- **16-bit** can store values from 0 to 65,535.
  - A total options of values = 2^16 = **65,536**.
- **32-bit** can store values from 0 to 4,294,967,295.
  - A total options of values = 2^32 = **4,294,967,296**.

BIT SIZE AND VALUES

- The programmer should decide the size of memory required to store a value.
  - What is the most efficient way to store values?
- Most programming languages have different data types.
  - To support varying values.
  - Can be considered as containers.
  - For example:
    - **Boolean** value: 1-bit.
    - **Character** reference: 8-bit
    - Short **integer**: 8-bit
    - **Integer** number: 16-bit

NEGATIVE NUMBERS

- Signed numbers have a bit reserved for the positive or negative sign.
- For example, 8-bit signed number stores a value up to 127.
    - But the same range of values from -128 to 127.

SCALE AND CONSTRAINTS

- Consider an Excel spreadsheet:
  - Each cell is assumed to store a 64-bit value (8 bytes).
  - Excel can allow for more than:
    - 1,048,576 rows
    - 16,384 columns
    - i.e., more than 17 billion cells (17,179,869,184 cells)
  - Such a spreadsheet requires 137,438,953,472 bytes (**128 GB**).
- **Can this be loaded/opened in a phone device?**

SCALE AND CONSTRAINTS (OVERFLOW)

- No room to store a value in a reserved container.
  - Can generate an error.
    - For example, the error by the calculator when the result does not fit the screen.
  - Sometimes something else happens:
    - **186 + 92 = 22** (not 278)

	1	0	1	1	1	0	1	0	186	
										+
	0	1	0	1	1	1	0	0	92	
1	0	0	0	1	0	1	1	0	22	

ENCODING TEXT

ASCII AND UNICODE

- **Can binary represent text?!**
- We can use a code for each letter and symbol.
  - **Encode:** represent the letter/symbol with its number.
  - **Decode:** use the number to get the letter/symbol.
- This encoding/decoding should be using the same code.
  - This is done with a **protocol**.

ASCII AND UNICODE

- Uppercase characters and lowercase characters
- Punctuation
- Common international (& Greek) characters,
- Special symbols
- Non-printed characters
    - Space, line return, backspace, and others.
- We have **256 characters**
    - We can encode these with **a single byte**.

ASCII AND UNICODE

The one-byte encoding system:

- is called ASCII
- American Standard Code for Information Interchange.
- Developed in the 1960s.

Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char	Dec	Hex	Oct	Char
0	0	0		32	20	40	[space]	64	40	100	@	96	60	140	`
1	1	1		33	21	41	!	65	41	101	A	97	61	141	a
2	2	2		34	22	42	"	66	42	102	B	98	62	142	b
3	3	3		35	23	43	#	67	43	103	C	99	63	143	c
4	4	4		36	24	44	$	68	44	104	D	100	64	144	d
5	5	5		37	25	45	%	69	45	105	E	101	65	145	e
6	6	6		38	26	46	&	70	46	106	F	102	66	146	f
7	7	7		39	27	47	'	71	47	107	G	103	67	147	g
8	8	10		40	28	50	(	72	48	110	H	104	68	150	h
9	9	11		41	29	51	)	73	49	111	I	105	69	151	i
10	A	12		42	2A	52	*	74	4A	112	J	106	6A	152	j
11	B	13		43	2B	53	+	75	4B	113	K	107	6B	153	k
12	C	14		44	2C	54	,	76	4C	114	L	108	6C	154	l
13	D	15		45	2D	55	-	77	4D	115	M	109	6D	155	m
14	E	16		46	2E	56	.	78	4E	116	N	110	6E	156	n
15	F	17		47	2F	57	/	79	4F	117	O	111	6F	157	o
16	10	20		48	30	60	0	80	50	120	P	112	70	160	p
17	11	21		49	31	61	1	81	51	121	Q	113	71	161	q
18	12	22		50	32	62	2	82	52	122	R	114	72	162	r
19	13	23		51	33	63	3	83	53	123	S	115	73	163	s
20	14	24		52	34	64	4	84	54	124	T	116	74	164	t
21	15	25		53	35	65	5	85	55	125	U	117	75	165	u
22	16	26		54	36	66	6	86	56	126	V	118	76	166	v
23	17	27		55	37	67	7	87	57	127	W	119	77	167	w
24	18	30		56	38	70	8	88	58	130	X	120	78	170	x
25	19	31		57	39	71	9	89	59	131	Y	121	79	171	y
26	1A	32		58	3A	72	:	90	5A	132	Z	122	7A	172	z
27	1B	33		59	3B	73	;	91	5B	133	[	123	7B	173	{
28	1C	34		60	3C	74	<	92	5C	134	\	124	7C	174	|
29	1D	35		61	3D	75	=	93	5D	135	]	125	7D	175	}
30	1E	36		62	3E	76	>	94	5E	136	^	126	7E	176	~
31	1F	37		63	3F	77	?	95	5F	137	_	127	7F	177	

ASCII AND UNICODE

Encode

Hello to you

72 101 108 108 111 116 111 121 111 117

01001000 01100101 01101100 01101100 01101111 01110100 01101111 01111001 01101111 01110101

Decode

01001000 01100101 01101100 01101100 01101111 01110100 01101111 01111001 01101111 01110101

72 101 108 108 111 116 111 121 111 117

Hello to you

05 MINUTES DISCUSSION

What about More letters, characters, and glyphs?

ASCII AND UNICODE

- **Unicode** was introduced to represent thousands of symbols from languages all around the world.
- Unicode uses multiple bytes to represents letters/symbols.

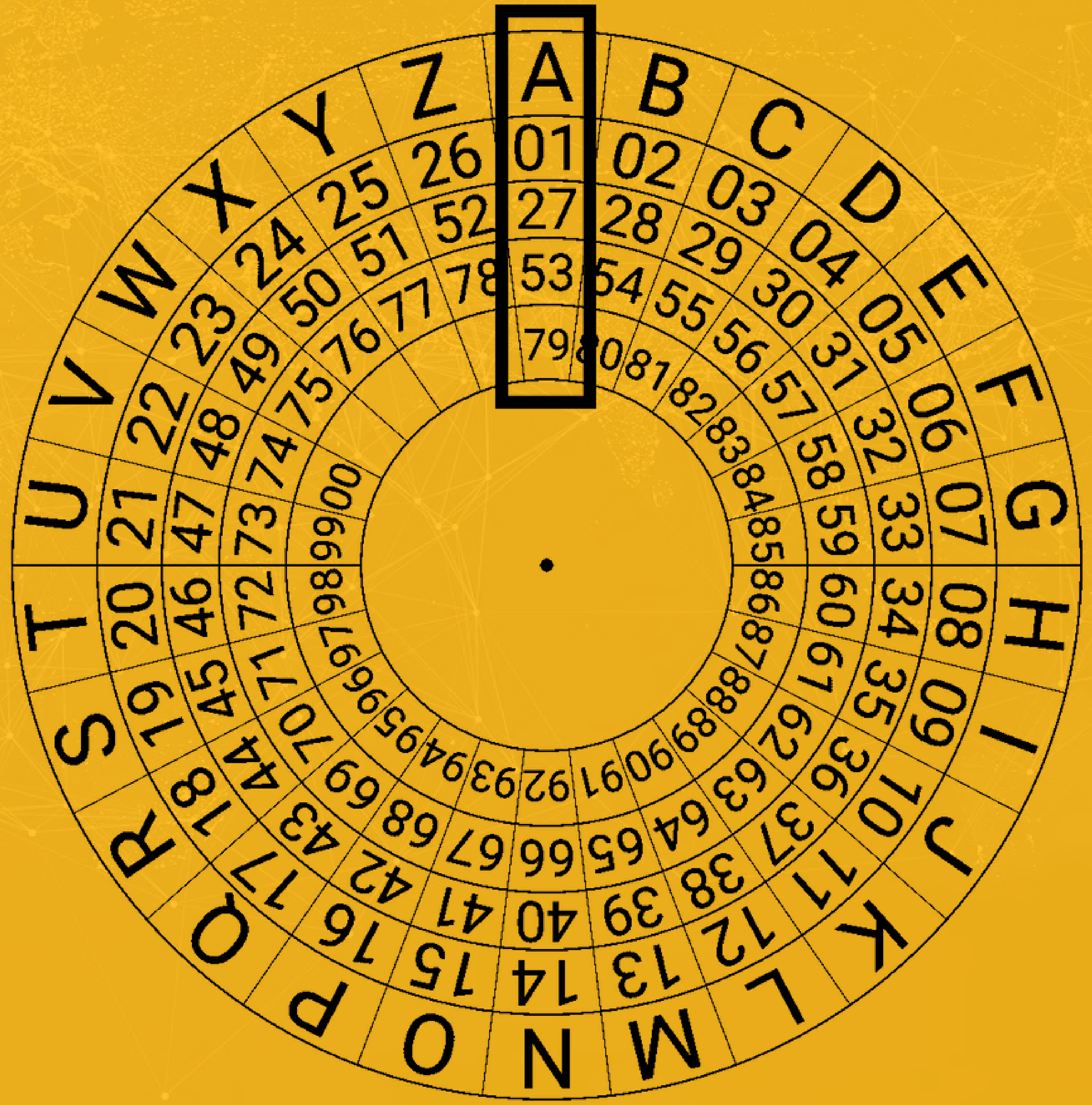
ASCII AND UNICODE

UTF-8 (8-bit Unicode Transformation Format)

- Compatible with ASCII
- Uses multiple planes (from 0 to 16 = 17)
- Uses a prefix U
- **Plane-0**
  - Uses a prefix U and then a four-digit hexadecimal number
    - Example: U0044 = D
    - It needs 16-bit (**2 bytes**)
  - Can represent more than **65,000 letters/symbols**.

WHAT ABOUT FORMAT?

Can we encode formatted texted?



ENCODING FORMAT

- We use different formatting features to add additional meaning to the text.

Text can be in **bold**, *italic*, <u>underline</u>, different color, and many other formats.

- **How does the encoding happen?**

ENCODING FORMAT

- Usually, it is done with a markup language such as HTML.
  - Tags are used to mark the start and end of a format.
  - The text editor/browser uses these tags to provide structure to the document.
- Tags are encoded with UTF-8.

Text can be in bold, <i>italic</i>, <u>underline</u>, and other format.

TEXT COMPRESSION

DATA MEASUREMENT

- The byte is the base form of measurement for data size.
- Data size is done by multiples of 1,024.
    - 1,024 bytes is a kilobyte (**KB**), can store (2-3 paragraphs).
    - 1,024 kilobytes is a megabyte (**MB**), can store (3-4 200-pages books).
    - 1,024 megabytes is a gigabyte (**GB**), can store (+4,000 books).
    - 1,024 gigabytes is a terabyte (**TB**), can store (+4,500,000 books).
    - Larger units:
        - Petabytes (**PB**), Exabytes (**EB**), Zettabytes (**ZB**), and Yottabytes (**YB**).

Large Files

Larger file size requires larger storage and transmission bandwidth and time.

Compression

Compression: makes the data space smaller while preserving the original data.

DATA COMPRESSION

DEFINITION

Data compression is the algorithmic problem of finding alternative, space-efficient encodings for a given data file.

- Compression requires a protocol defining the encoding and decoding method.
  - **Lossless:** recovered information from the compressed form is the same as the original.
    - Zip
  - **Lossy:** recovered data differs (insignificantly) from the original.
    - jpeg compression

TEXT COMPRESSION

- Text compression is done using lossless compression
    - **Codewords** and **Dictionaries** are used to eliminate redundant information.

TEXT COMPRESSION (EXAMPLE)

DICTIONARY

1. ASK
2. NOT
3. WHAT
4. YOUR
5. COUNTRY
6. CAN
7. DO
8. FOR
9. YOU

ORIGINAL TEXT

ASK NOT WHAT YOUR COUNTRY CAN DO FOR YOU
ASKWHAT YOU CAN DO FOR YOUR COUNTRY

ENCODED TEXT

1 2 3 4 5 6 7 8 9 1 3 9 6 7 8 4 5

TEXT COMPRESSION

- Multiple passes can be done to further compress the content.
  - To save space and time.
  - To allow for better analysis

ENDCODING IMAGES

IMAGES

- Images are based on individual pixels.
  - Bitmaps
  - Matrix of pixels
    - The hight of an image defines the number of rows.
    - The width of an image defines the number of columns.
  - Each pixel holds a color.
    - **RGB (#000000)**
    - Each color (red, green, and blue) is given value .
      - A color value is a 2-digit hexadecimal number (**1 byte**).
    - These color values are encoded into binary.

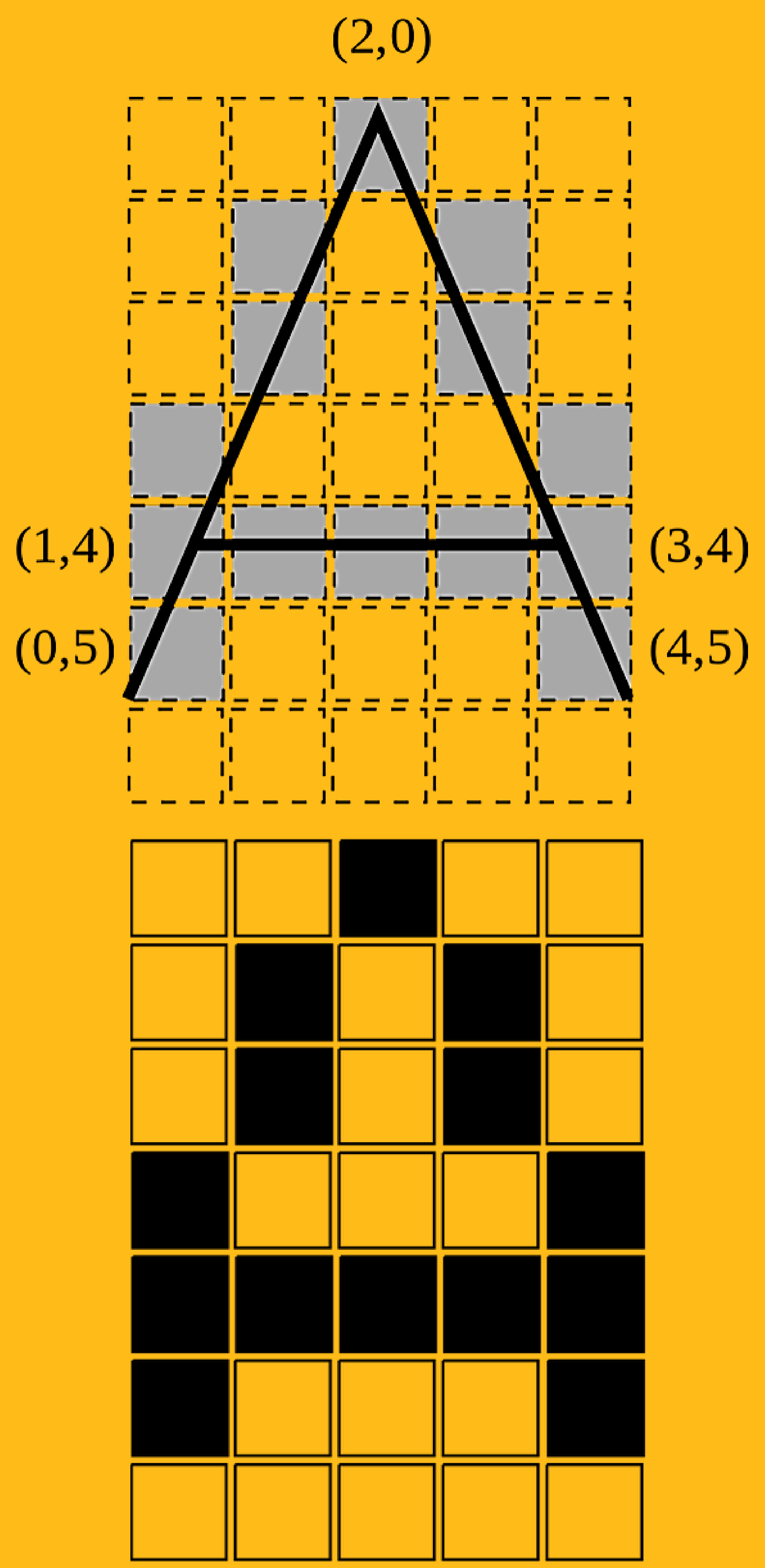


IMAGE RESOLUTION

- **HD RESOLUTION IMAGE**
  - 1920 x 1080 = 2,073,600 pixels.
  - 2,073,600 x 3 bytes = 6,220,800 bytes
  - 5.9326 megabyte

- **4K RESOLUTION IMAGE**
  - Four times the size of an HD image.
  - 3,840 x 2,160, = 8,294,400 pixels.
  - 8,294,400 x 3 = 24,883,200 bytes
  - 23.730 megabytes

IMAGE FILE

IMAGE FILES HAVE MORE INFORMATION THAN THE ACTUAL PIXEL VALUES.

- The origin
- Camera
- Date
- Owner
- etc.

SO THE FILE SIZE IS SLIGHTLY LARGER THAN THE IMAGE PIXEL INFORMATION.

VIDEO FILES

- **HD RESOLUTION VIDEO**
  - 30 frames per second (could be more).
  - 30 x 6,220,800 bytes (HD) = 186,624,000 bytes/second.
  - 186,624,000 x 60 seconds = 11,197,440,000 bytes/minute.
  - **10.428 gigabyte/minute.**

- **HOW ABOUT 4K VIDEO?**
  - more than **41 gigabyte/minute.**
  - That is the size **without the audio** or other information.

HOW TO ENCODE IMAGES?

VECTORS

BITMAPS

BITMAPS

- The most common image formats.
    - Display images on a computer monitor, printers, scanners, and similar devices.
- Bitmaps contain pixels.
- Example of file formats:
    - GIF, JPEG, PNG, TIFF, XBM, BMP, and PCX.

VECTORS

- Vector images are constructed using mathematical formulas.
  - Describing **shapes**, **colors**, and **placement**.
- Vector images consist of <u>shapes</u>, <u>curves</u>, <u>lines</u>, and <u>text</u>.
- Example of file formats:
  - CMX (Corel Metafile Exchange Image), **SVG** (Scalable Vector Graphics), CGM (Computer Graphics Metafile), DXF, and WMF (Windows Metafile).
- Resolution-independent.
- Maximum quality regardless of scale.

BITMAPS AND VECTORS

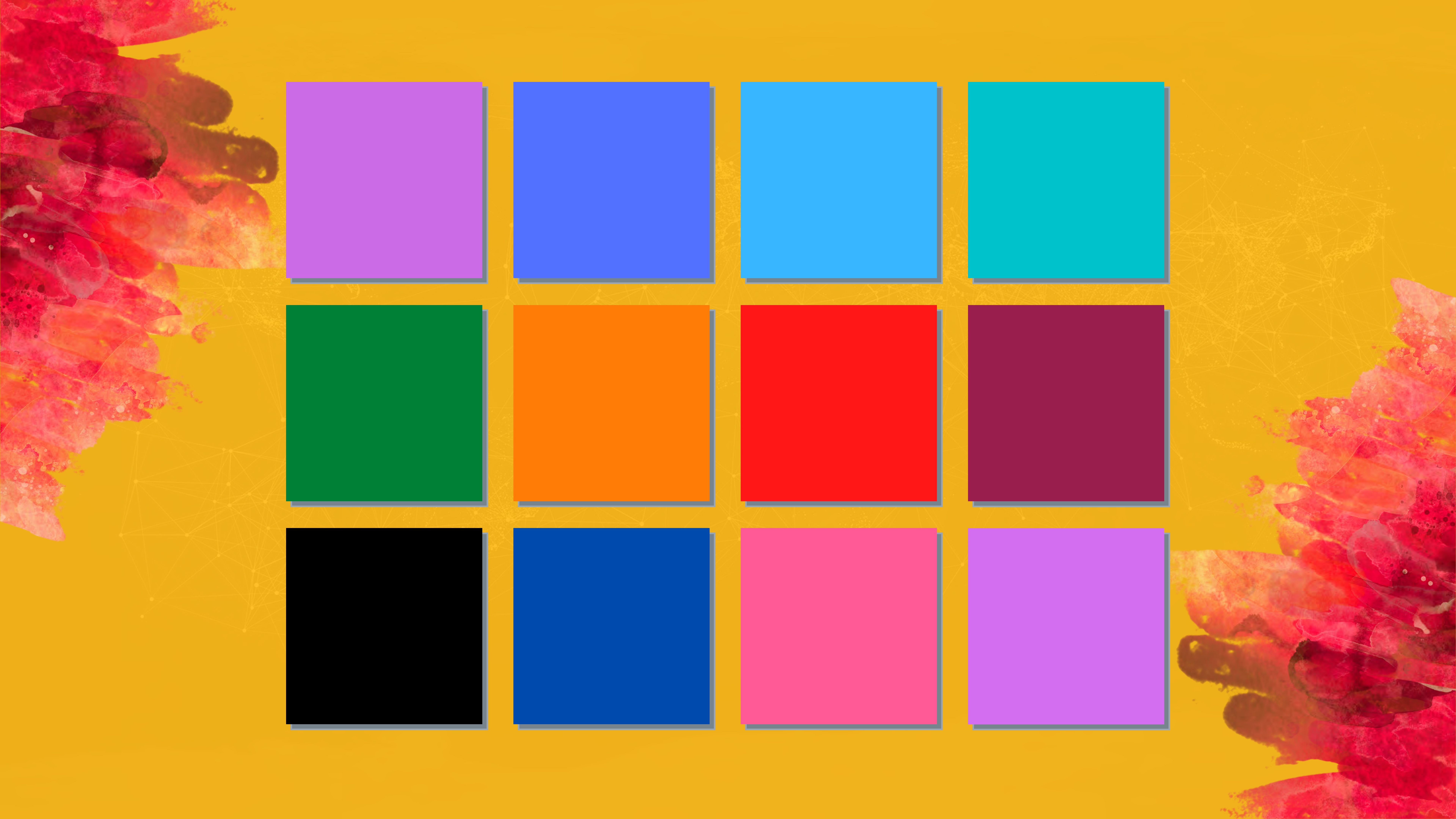
Web browsers and applications work with both formats.

IMAGE SIZE AND COLOR VARIATIONS

WHAT CAN WE DO TO REDUCE THE SIZE OF IMAGES?

Think of the number of colors that we can choose from!

- Consider RGB 3-byte color range (e.g. #FFFFFF)



R: 205

G: 108

B: 230

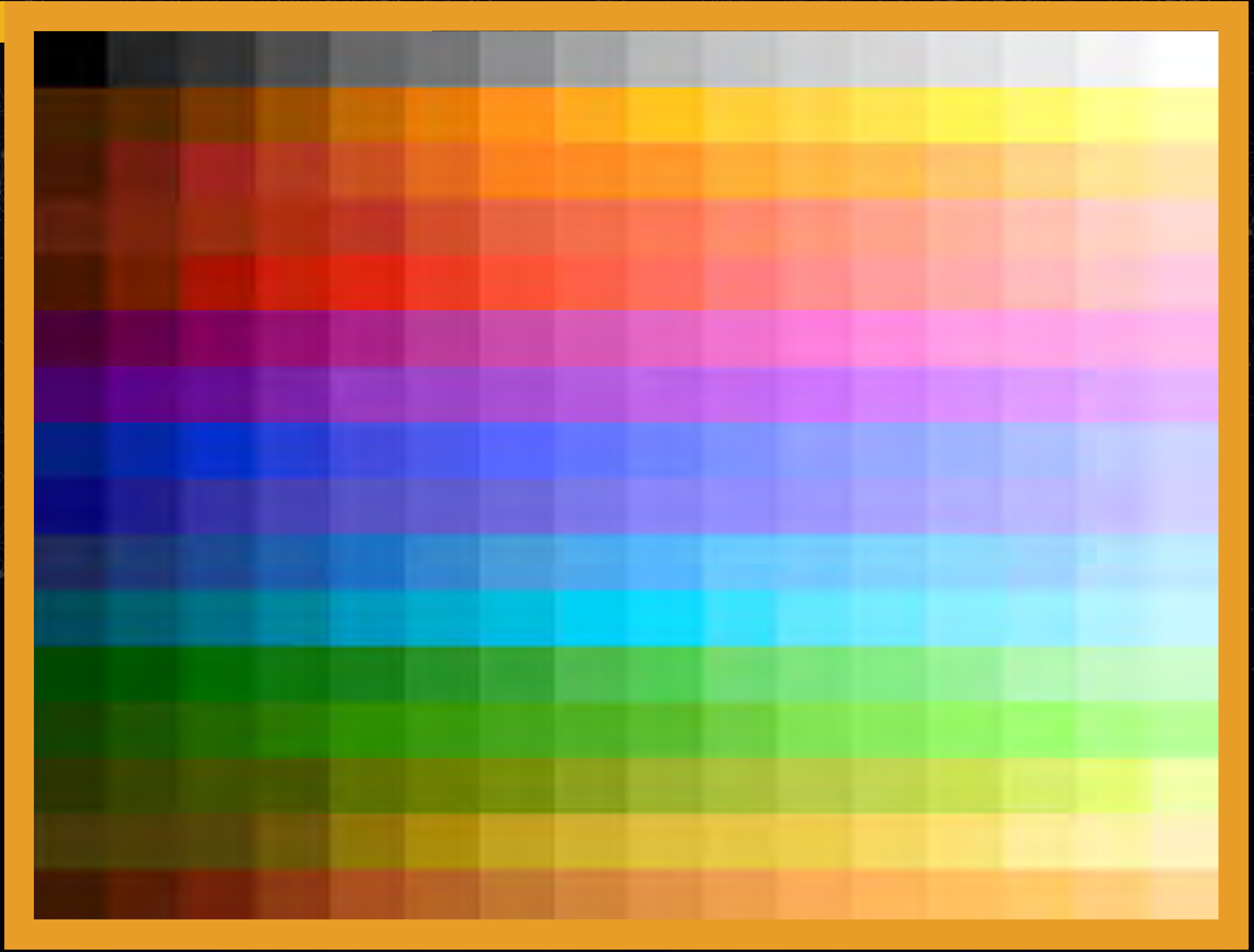
R: 210

G: 110

B: 239

GIF

- Reduce the number of color variations
- GIF uses a total number of **256 colors**.
  - Select from **24-bit color space**.
- This way the image can be compressed with a lossless compression.



JPEG

- JPEG Uses **lossy compression** to minimize the size of the image.
- Once compressed, the image cannot return to its original format.
- There are different levels for compression
  - From no loss to significant loss.

More compression

PNG

- PNG: Portable Network Graphics
- Improved version of the GIF format.
  - Pixels can contain 24-bit RGB color values or 32-bit values.
  - PNG also includes transparency.

OTHER IMAGE FORMAT

- Other formats and color systems are used for other purposes.
  - For example, print and photography.
  - **CMYK** (Cyan, Magenta, Yellow, and Black) are the primary colors for print.
  - Camera sensors capture light information in **RAW** format.
    - A program is used to translate RAW data into pixels.
- Image files can vary in type, context, and format.

END OF MODULE 1